

Machine Learning and Creative Practice

MusAI Workshop 2026

AI Music: Critical Interdisciplinary Perspectives, Beyond Ethics

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abstract

What kinds of technical and artistic innovation does ML enable? What is the relationship between them and how are they shaped by aesthetic concerns? Can artistic work with ML serve as a source of critique, informed by humanities scholarship? We examine contemporary and critical practice in musical composition with ML.

readings

core readings

- Einbond, A. 'Magic of Machine Learning: Interdisciplinary Composition and Collaboration in *Prestidigitation*' (forthcoming 2026)
- Gioti, A. 'Listening like an Algorithm: Ethnography of a Machine Listening System' (forthcoming 2026)

further readings

- Gioti, Artemi-Maria, Aaron Einbond, and Georgina Born. 'Composing the Assemblage: Probing Aesthetic and Technical Dimensions of Artistic Creation with Machine Learning'. *Computer Music Journal* 46, no. 4 (2022): 62–80. https://doi.org/10.1162/comj_a_00658.
- Born, G. (2005). On Musical Mediation: Ontology, Technology and Creativity. *Twentieth-Century Music*, 2(1), 7–36.
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research questions

Gioti, Einbond, and Born (2022)

1. How does ML shape aesthetic choices in distinctive compositional practices, and how is it shaped in turn by aesthetic commitments?
2. How can critical themes from humanities scholarship inform artistic research in composition using ML?
3. In what ways does the interaction with ML algorithms as part of the compositional process differ from that with other music technology tools?

research questions

- We address these research questions through two case studies undertaken by composer-researchers creating new compositions with ML
- Auto-ethnography, a method increasingly common in artistic creation and music research (Findlay-Walsh 2018, 121-122, Magnusson 2011, 609-610), is used here as a means of deepening critical reflection drawing on the ethnographic paradigm of Born (Born 1995, 2004, Born & Barry 2013)
- By putting ethnography in dialogue with theory, we adopt an epistemological stance that Born (2010, 2022) terms “post-positivist empiricism” according to which ethnography can be “a subtle tool for the application and the amendment of theory” (Born 2010, 197–8)

research questions

- We examine not only narrowly defined ML algorithms but also the broader assemblage (Born 2005, 2011), in which each musical work is embedded: instruments, microphones, loudspeakers, audio descriptors, code, performers, and audience.
- Emphasize the “aesthetic situatedness” (Snape and Born 2022, p. 223) of composers’ uses of ML as they participate in the broader compositional assemblage.
- Past work has addressed aesthetic engagement with interactive music technology systems not involving ML as understood here (Di Scipio 1998; Lewis 1999; Impett 2000)
- or ML in instrument building and performance practice (Fiebrink and Sonami 2020)
- but less work has addressed composition, our focus.

research questions

- critical themes from humanities scholarship:
 - *material engagement* (Malafouris 2004, 2013; Knappett and Malafouris 2008; Jones and Boivin 2010)
 - *critical data studies* (Boyd and Crawford 2012; Markham 2013; Crawford and Paglen 2021; Poirier 2021)
- disciplinary context: creative practice research
 - ML assemblage as something not just “used” but “composed”
 - AI not only *trained* on creative content—AI *is* the content

material engagement



Fig. 2.1 At the potter's wheel (Malafouris 2008, 20)

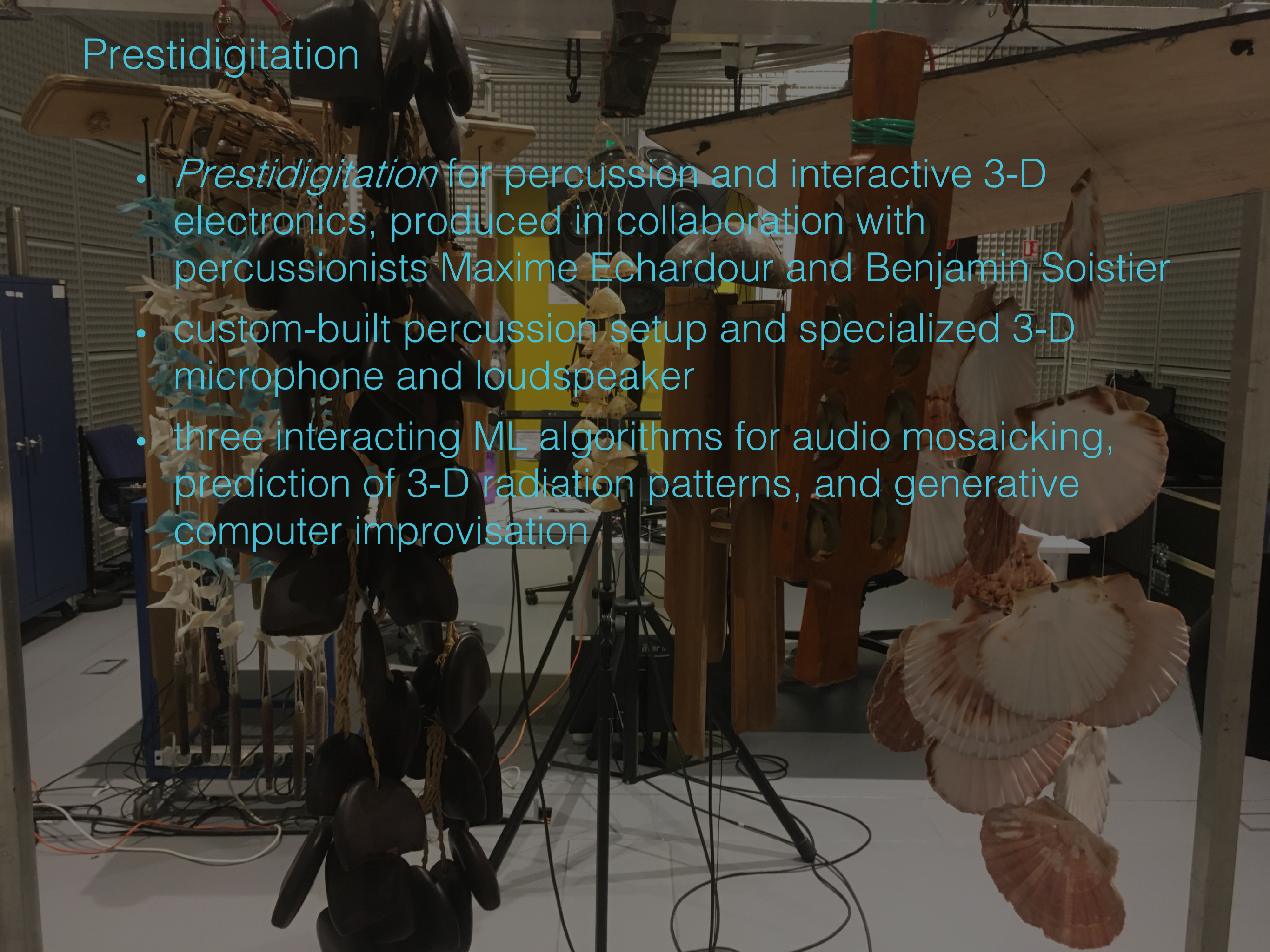
material engagement

- “At one moment, movement is effortless and feels like happening to the potter rather than being done by the potter as if totally absorbed into the micro-structure of clay. At another moment, the potter is clearly conscious of moving clay around and shaping it, directing the flow of the clay and struggling to control the act and handle the clay.” (Malafouris 2008, 34)
- “Agency is a property or possession neither of humans nor of nonhumans. Agency is the relational and emergent product of material engagement. It is not something given but something to become realised” (Malafouris 2008, 34)
- Like potter and clay, the performer, the computer, and I all engage in a gradual push-and-pull where we each, human and nonhuman, listen, react, then make iterative adjustments to reshape our interactions.

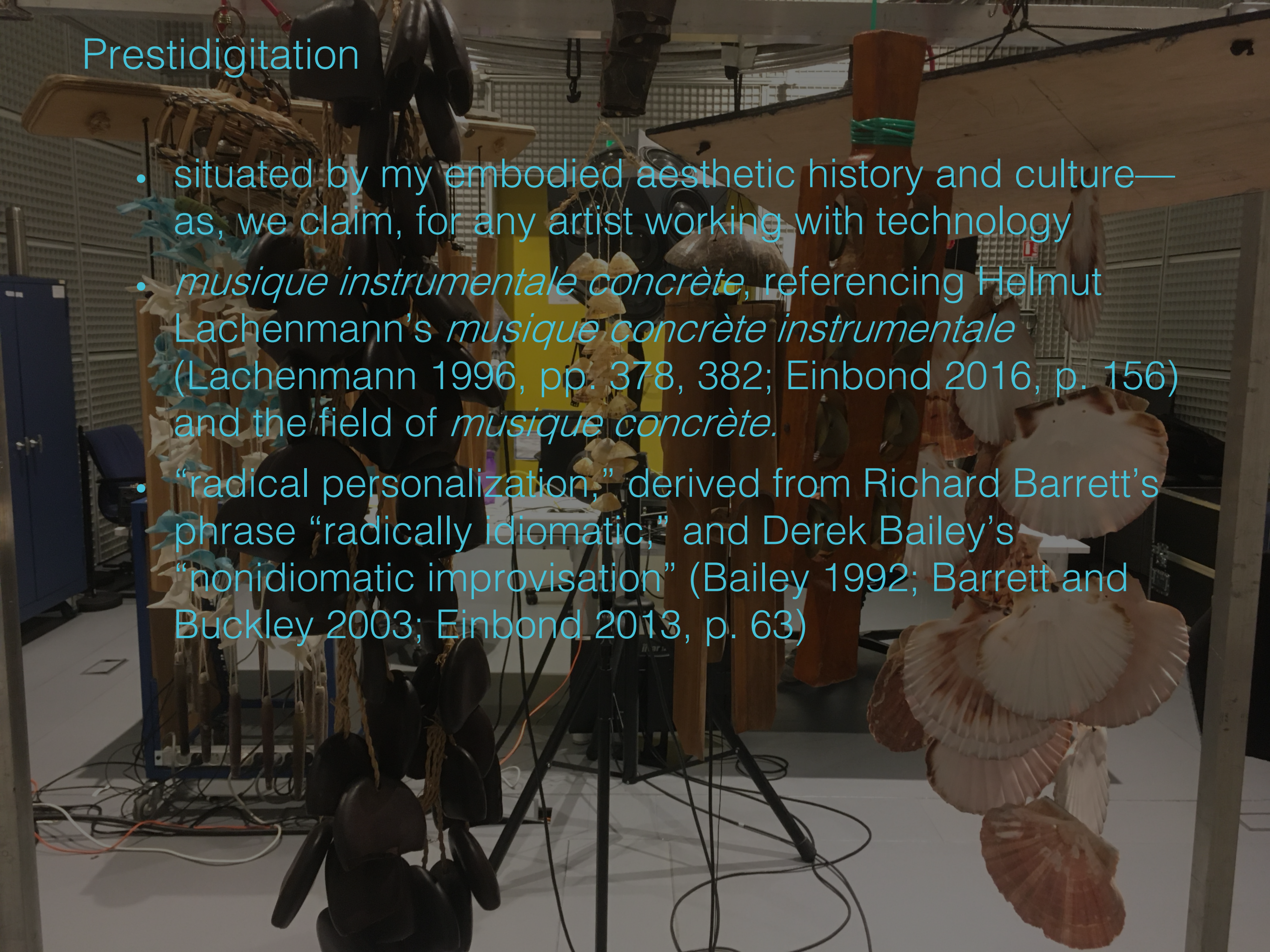
Fig. 2.1 At the potter's wheel

Prestidigitation

- *Prestidigitation* for percussion and interactive 3-D electronics, produced in collaboration with percussionists Maxime Echardour and Benjamin Soistier
- custom-built percussion setup and specialized 3-D microphone and loudspeaker
- three interacting ML algorithms for audio mosaicking, prediction of 3-D radiation patterns, and generative computer improvisation



Prestidigitation



- situated by my embodied aesthetic history and culture—as, we claim, for any artist working with technology
- *musique instrumentale concrète*, referencing Helmut Lachenmann's *musique concrète instrumentale* (Lachenmann 1996, pp. 378, 382; Einbond 2016, p. 156) and the field of *musique concrète*.
- “radical personalization,” derived from Richard Barrett’s phrase “radically idiomatic,” and Derek Bailey’s “nonidiomatic improvisation” (Bailey 1992; Barrett and Buckley 2003; Einbond 2013, p. 63)

Prestidigitation

- Gaussian Mixture Mode (GMM) trained on a database of 3-D radiation patterns of acoustic musical instruments (Weinzierl et al. 2017)
- k-nearest neighbors (k-NN) concatenates samples of percussionist-collaborator Maxime Echardour in audio mosaic
- audio oracle (AO) trained in real-time to perform “computer noise improvisation”
- Excerpt from performance by Benjamin Soistier
Performing Critical AI II, IKLECTIK, London, 2022
<https://youtu.be/zi9COmpU9ms?feature=shared&t=1069>

Prestidigitation



4

3'00"

3'05"

3'10"

R

L

Elec.

mount grelots, take cowbells

12 play oracle

transcribed and recomposed human improvisation

3'15"

3'20"

3'25"

R

L

Elec.

grelots

cowbells

13 pause oracle, continue recording

3'30"

3'35"

3'40"

R

L

Elec.

mount cowbells

improvise with indicated instruments in response to electronics

14 play oracle

computer improvisation

human improvisation

3'45"

3'50"

3'55"

R

L

Elec.

take chacha shaker

chacha shaker

15 pause oracle, continue recording

magic of ML

- Forthcoming (2026)
- “The defining feature of 'magic' as an ideal technology is that it is 'costless' in terms of the kind of drudgery, hazards and investments which actual technical activity inevitably requires. Production 'by magic' is production minus the disadvantageous side-effects, such as struggle, effort, etc.” (Gell 1988, p. 9)

magic of ML

1. ML conceals layers of human labor that crystalize each collaborator's aesthetic background in the process.
2. algorithmic complexity and musical richness are not necessarily correlated: simple ML algorithms and “small data” can be creatively stimulating.
3. Compared to other music technological assemblages, ML differs in the temporality and ontology of music-making due to the unique affordances of training data.
4. ML tools are contingent on human, social, and institutional forces that can cultivate “permeable interdisciplinarity”, blurring and redefining the roles of composer, performer, producer, and researcher.

magic of ML

- *Prestidigitation II* for percussion solo, five instruments, and 3-D electronics:

<https://youtu.be/eL6TooEwtHY?feature=shared>





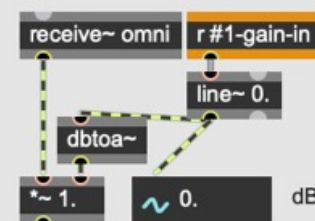




magic of ML

- Small data: three interacting ML models
- Gaussian Mixture Mode (GMM) trained on a database of 3-D radiation patterns of acoustic musical instruments (Weinzierl et al. 2017)
- k-nearest neighbors (k-NN) concatenates samples of percussionist-collaborator Echardour in an “audio mosaic”
- audio oracle (AO) trained in real-time to perform “computer noise improvisation”

The screenshot shows a Pure Data patch for 3D rotation. It features three rotary knobs for 'yaw', 'pitch', and 'roll'. The yaw knob is set to -180.00 d, while pitch and roll are at 0.00. Each knob is connected to a message box: /yaw \$1, /pitch \$1, and /roll \$1. These message boxes are connected to a large object: spat5.hoa.rotate~ @mc 1 @dimension 3 @order 3. On the left, a line~ 0. object is connected to a dbtoa~ object, which is then connected to a mc.*~ 0. object. The final output is mc.send~ cat-spat 16.



live control

```
go ircamdescriptor-onseg
```

- p target-analysis
-

keys

p select-key	p select-buffer
--------------	-----------------

classify
p gmm-classify  Double_bass
_modern

synthesis		absolute + relative	
duration	▶ 0.	▶ 1.	
durationvar	▶ 0.	▶ 0.	
attack	▶ 5.	▶ 0.	
release	▶ 5.	▶ 0.	
offset	▶ -5.		
positionvar	▶ 0.		
level	▶ 10.		
levelvar	▶ 0.		
resampling	▶ 0.		
resamplingvar	▶ 0.		
outputgains			▶
-0.3125 -0.076869 0.005072 -0.193517 0.01			

p presets

p presets2

p OSC-router

view
p imubu

audio oracle

start concat

pyoracle-osc
#1

learn

improve

0 0

draw length next state next

The screenshot shows the 'pattr' (pattern reader) interface. It features a title bar 'pattr' and a main area with several controls:

- write** and **read** buttons on the left.
- store** and **call** buttons in the center.
- clear** button on the right.
- autopattr** button with a play icon on the far right.
- storagewindow** dropdown menu.
- A status bar at the bottom displaying: `pattrstorage storage @autorestore 0 @changemode 1`

magic of ML

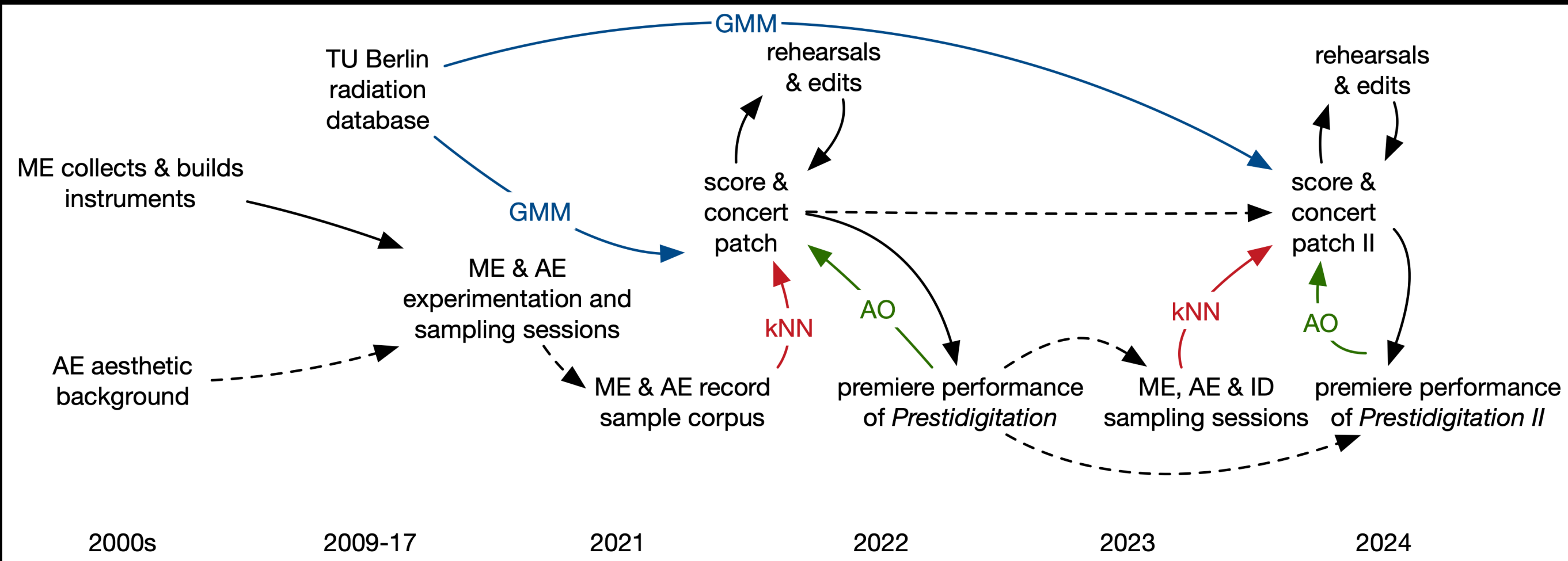
- aesthetic background and social contexts:
- small data
- “interdisciplinarity in one person” (Barry and Born, 2013, p. 29)
- “*agonistic-antagonistic* mode” of interdisciplinarity (Barry & Born, 2013, p. 12, original italics)
- “permeable interdisciplinarity”
- contrasts and commonalities with *Rationalizing Culture* (Born 1995)
- IRCAM hybrid research group *Music ex machina* 2025-28

magic of ML

- “work concept” (Goehr, 1992)
- Born challenges Goehr’s “reluctance to pursue the significance of technologies of music production and reproduction for the shifting ontology of contemporary music. [...] I want to show the implications of the new technologies and associated practices for the work concept, which, I will propose, is more destabilized than Goehr allows.” (2005, p. 10)
- “Distributed across space, time and persons, music can become an object of recurrent decomposition, composition and re-composition by a series of creative agents. We need a new term for this capacity: I suggest relayed creativity.” (2005, p. 26)
- ML augments this more than ever

magic of ML

- Protections and retentions in *Prestidigitatation I-II*



Aaron Einbond

PRESTIDIGITATION II

(2023)

für Solo-Schlagzeug, Bassflöte, Bassklarinette, Violine, Viola, Violoncello und dreidimensionale Elektronik
for percussion solo, bass flute, bass clarinet, violin, viola, violoncello, and three-dimensional electronics

Partitur / Score



eg 3089LM

253 $\frac{5}{4}$ Z

B.Fl.

B.Cl.

R

L

F.D.

Elec.

improvise in response to electronics

m255 pause oracle

The musical score is written for six staves: B.Fl., B.Cl., R, L, F.D., and Elec. The time signature is 5/4. Measure 253 begins with a treble clef, a 5/4 time signature, and a 'Z' in a box. The Elec. staff has a green box from measure 253 to 255. Measure 255 has a circled 'm255' and a 'pause oracle' instruction. A blue box highlights the improvisation instruction for measures 254-255.

256

B.Fl.

B.Cl.

R

L

F.D.

Elec

m256 play oracle 2 (ensemble) to IKO

m257 pause oracle

m258 play oracle 1 (perc) to IKO

improvise with all chimes
in response to electronics
and bfl.

uncreative writing

- precedent for “relayed creativitiy” in dialogue with technological assemblage:
- Kenneth Goldsmith,
Uncreative Writing (2011)

Uncreative Writing

Kenneth Goldsmith

uncreative writing

- “A final ontological shift concerns the materiality of digitized music, and is clarified by comparison with earlier forms of sonic representation, appropriation and remediation associated with recorded music. [...] What happens when we hold Goldsmith’s ideas up against appropriation and intertextuality in sound montage and sampling?
- “AI brings a more drastic transformation of sonic materials. [...] there’s not just a weakening – as with sampling – but an **obliteration** of sources. [...] I want to suggest that the data-fied sonic materials underlying generative AI, then, should be conceptualised as an **aggregate**. (Born, 2024)

uncreative writing

- “For the past several years, I’ve taught a class at the University of Pennsylvania called ‘Uncreative Writing.’ In it, students are penalized for showing any shred of originality and creativity. [...] We retype documents and transcribe audio clips.” (2011, 8)
- “The secret: the suppression of self-expression is impossible. Even when we do something as seemingly “uncreative” as retyping a few pages, we express ourselves in a variety of ways. The act of choosing and reframing tells us [...] much about ourselves” (2011, 9)
- “A class of 15 students will bring in 15 unique and individual pieces proving that the act of transcription is an idiosyncratic and personal endeavor—how I hear and transcribe something will be different from you.” (2013, 174)

questions

- In what ways does the interaction with ML algorithms as part of the creative process differ from that with other elements of the music technology assemblage?
- What does “uncreative writing” with ML have in common, or not, with other forms of intertextuality (cut-up, sampling)
- How does ML contribute to challenge concepts of the musical work and authorship, and are these challenges new?

acknowledgements

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European Research Council
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